Supply Chain and Inventory Analytics Report

Procurement, inventory, shipping and demand analysis

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1. Executive Summary

This report analyses procurement, inventory, shipping and demand data for a supply chain with 4 suppliers, 4 raw materials and 3 warehouses. Ghana Cocoa Beans drives 57.7% of procurement spend, and Ghana Cocoa Beans makes up 40.5% of stock value. About 25.1% of shipments are delayed across the whole network and 22.9% of purchase orders are cancelled. Demand is steady, so the main opportunities are stock rebalancing, supplier management and better delivery-date tracking.

Key figures

Metric	Value
Units in stock	54,643
Stock value	377,284
Procurement spend (PO lines)	29,192,118
Suppliers / Materials / Warehouses	4 / 4 / 3
Purchase orders	1,062
PO cancellation rate	22.9%
Sales orders	4,339
Units demanded	455,752
Shipments marked Delayed	25.1%

2. Dataset Overview

Data was generated with the Excelx Supply Chain and Logistics Data Generator in its slightly messy setting. It contains related tables covering procurement, inventory, transport and order fulfillment. The cleaned tables are summarised below.

Table	Rows	Columns
carriers	4	2
fulfillment_status	4,446	4
purchase_order_lines	2,714	6
purchase_orders	1,062	6
raw_materials	4	4
sales_orders	4,339	6
shipment_tracking	4,557	6
stock_movements	526	6
supplier_performance	4	5
suppliers	4	3
warehouse_inventory	12	5
warehouses	3	3

3. Data Cleaning Summary

Table	Raw rows	Clean rows	Removed	% removed
carriers	4	4	0	0.0%
fulfillment_status	5,000	4,446	554	11.1%
purchase_order_lines	3,141	2,714	427	13.6%
purchase_orders	1,250	1,062	188	15.0%
raw_materials	4	4	0	0.0%
sales_orders	5,000	4,339	661	13.2%
shipment_tracking	5,000	4,557	443	8.9%
stock_movements	712	526	186	26.1%
supplier_performance	4	4	0	0.0%
suppliers	4	4	0	0.0%
warehouse_inventory	12	12	0	0.0%
warehouses	3	3	0	0.0%

Actions taken

- Removed stray spaces from text fields and converted date columns to date type.
- Removed exact duplicate rows and repeated IDs.
- Dropped rows missing join keys (order, shipment, PO, material, supplier or warehouse IDs).
- Filled missing text categories with 'Unknown'; recovered product names from product IDs; restored a missing supplier from the performance table; filled missing unit costs with the material median.
- Dropped rows missing quantity or the main date, and rows dated in the future.
- Added LineTotal = Quantity x UnitCost; renamed TotalCost to ReportedTotalCost because it did not match the order lines.
- Left shipment and expected delivery dates blank where missing rather than inventing delays.

4. Statistical Summary

Measure	Count	Mean	Std	Min	Median	Max
PO line quantity	2,714	1,258.9	427.0	500.0	1,254.5	2,000.0
PO unit cost	2,714	8.5	7.3	0.5	5.1	20.5
PO line total	2,714	10,756.1	10,422.7	250.0	8,109.0	41,000.0
Sales order quantity	4,339	105.0	54.8	10.0	105.0	200.0
Inbound receipt quantity	526	1,244.4	441.5	510.0	1,235.0	1,997.0
Stock on hand per position	12	4,553.6	3,266.0	625.0	3,174.5	9,828.0

5. Visualizations

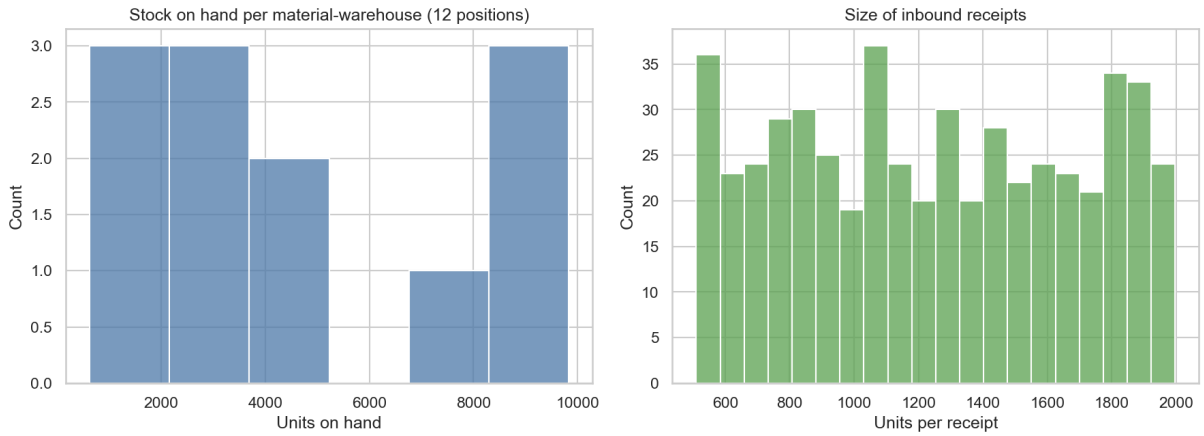


Figure 1: Inventory distribution

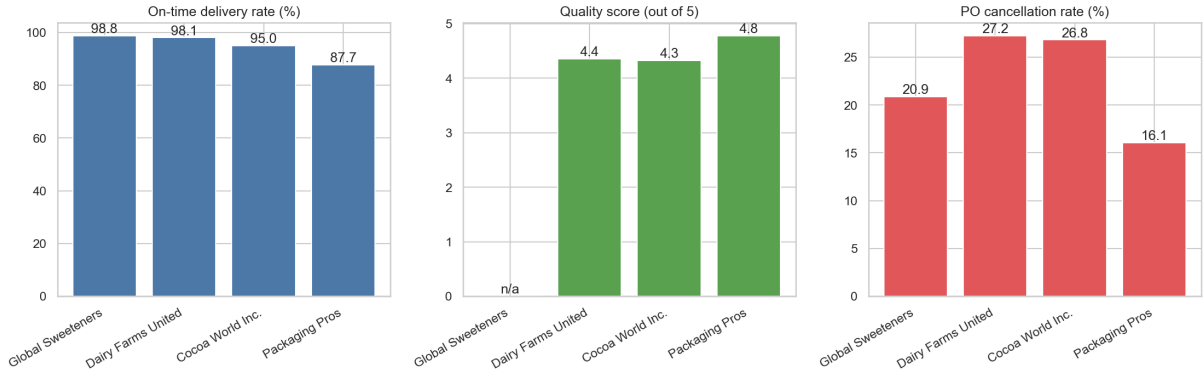


Figure 2: Supplier performance

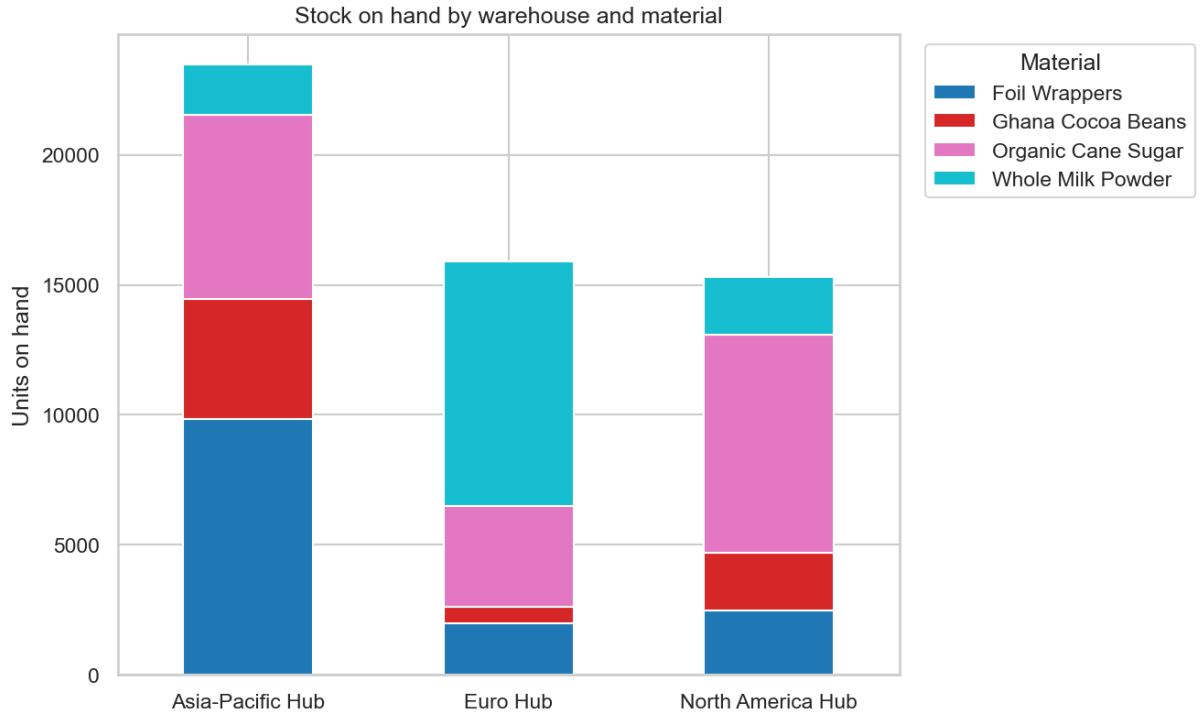


Figure 3: Warehouse stock by material

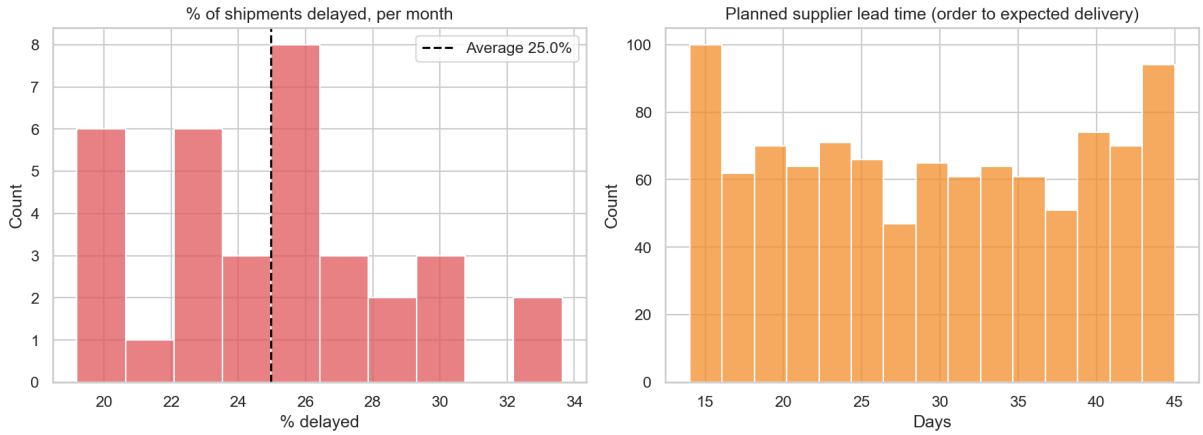


Figure 4: Shipping delay and planned lead time

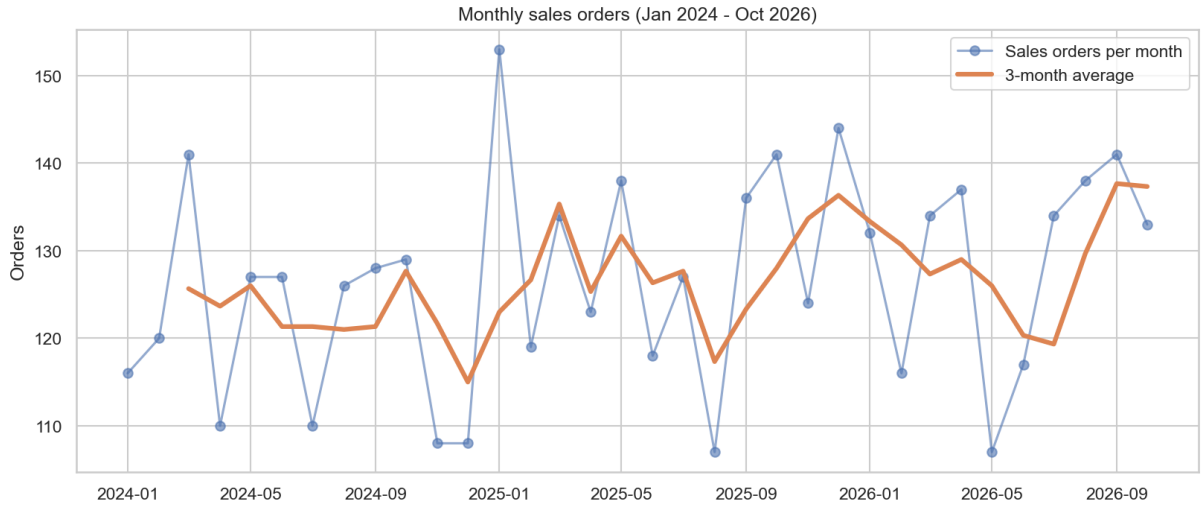


Figure 5: Monthly sales orders

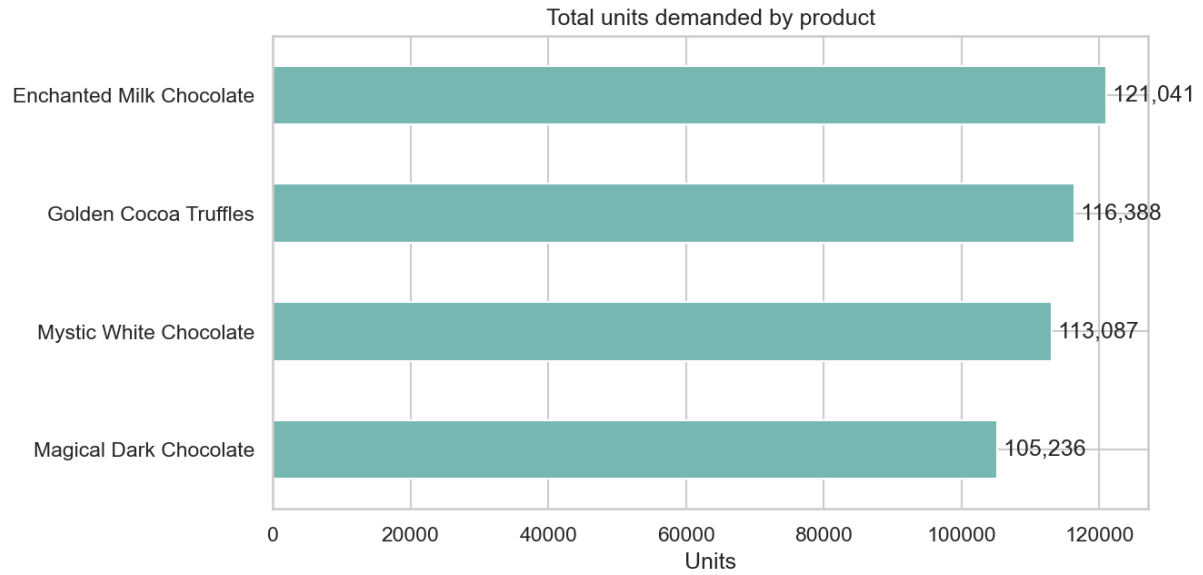


Figure 6: Product demand

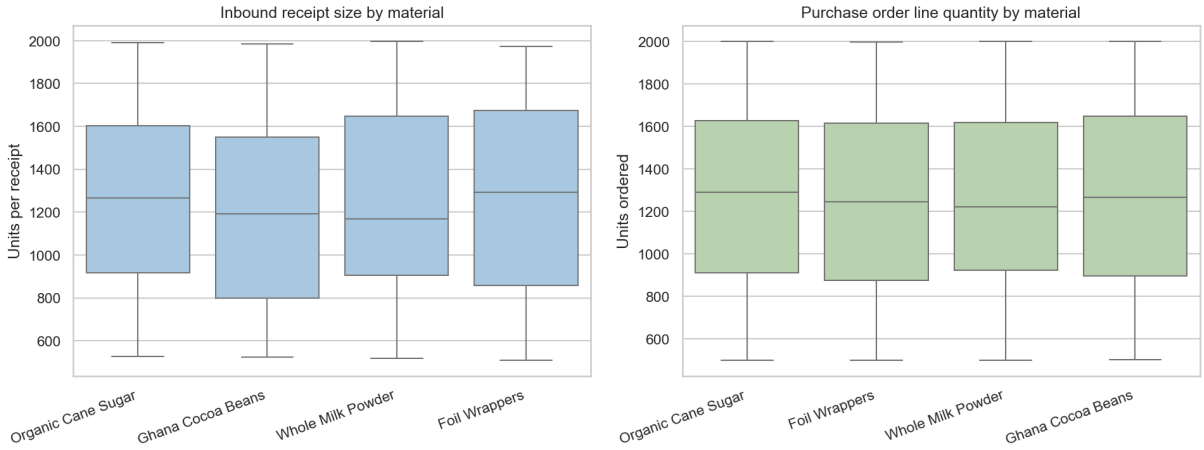


Figure 7: Receipt and order quantities by material

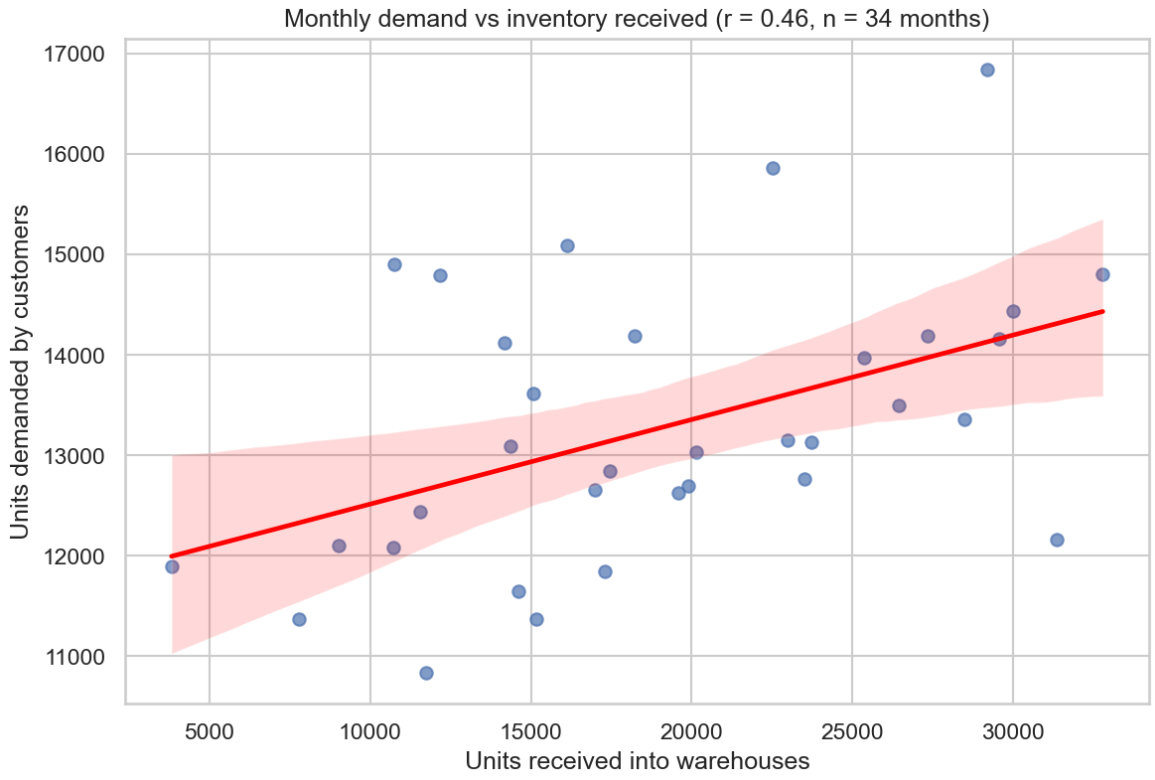


Figure 8: Demand vs inventory received

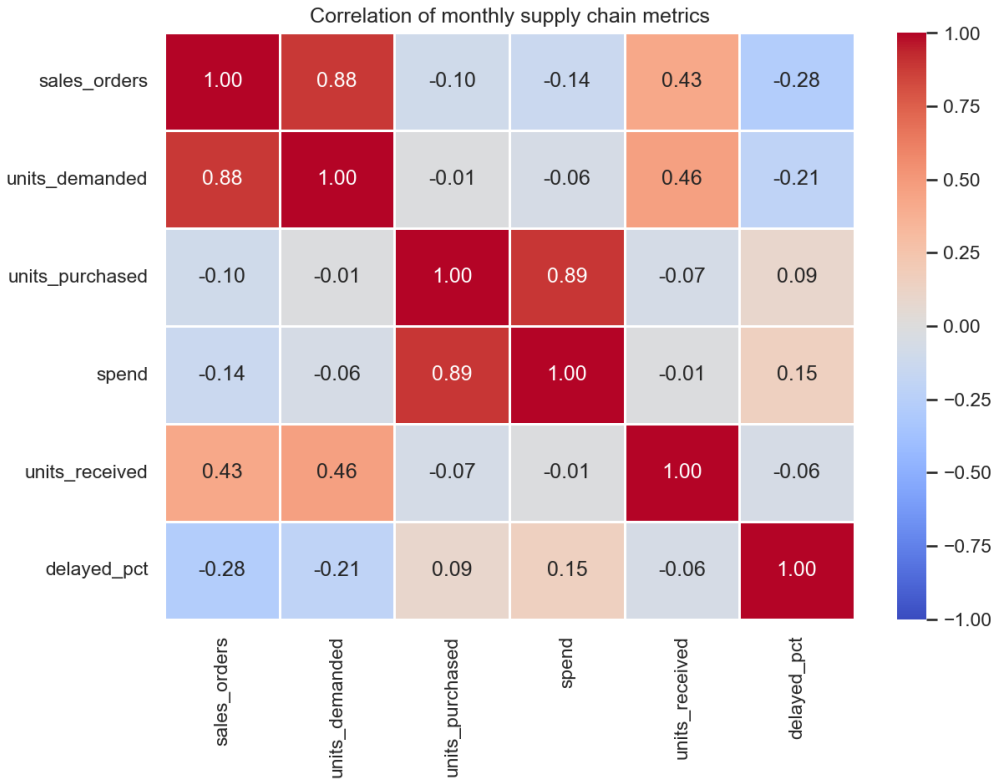


Figure 9: Correlation heatmap

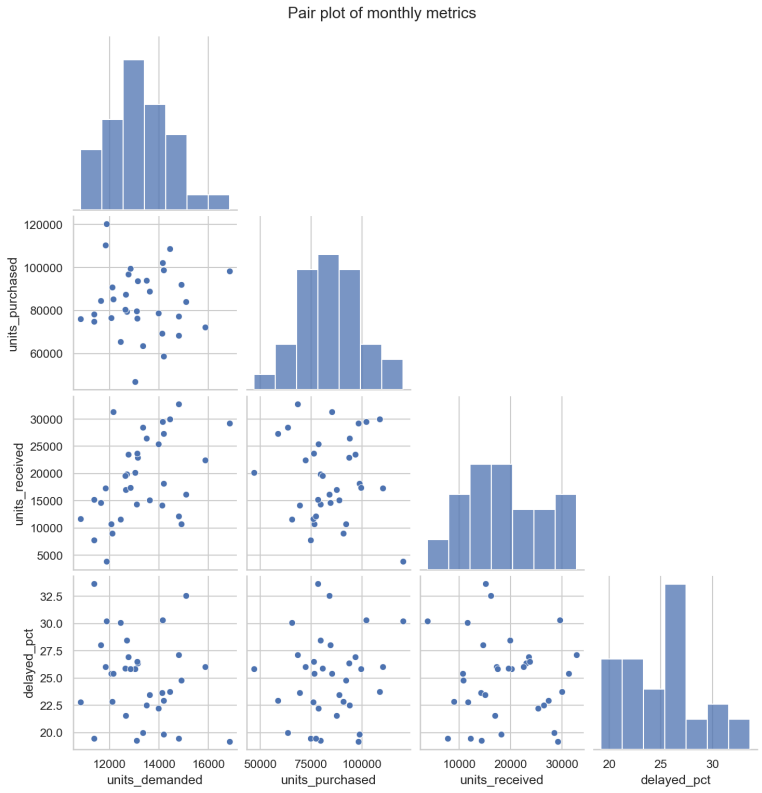


Figure 10: Pair plot

6. Business Insights

1. Procurement spend is concentrated in one material

Ghana Cocoa Beans accounts for 57.7% of total procurement spend.

Recommendation: Focus negotiation, dual sourcing and price monitoring on this material first.

2. A high-value material is thin in some warehouses

Ghana Cocoa Beans is 40.5% of stock value but only 13.6% of units. The lowest stock position is Ghana Cocoa Beans at Euro Hub (625 units).

Recommendation: Set reorder points for the lowest positions and rebalance stock between warehouses.

3. Stock is unevenly spread across warehouses

Asia-Pacific Hub holds 43.0% of all units on hand, while North America Hub holds 28.0%.

Recommendation: Review inter-warehouse transfers so stock sits close to where it is needed.

4. Suppliers trade off reliability, speed and quality

Packaging Pros has the lowest on-time rate (87.7%), while Global Sweeteners has the highest (98.8%). Global Sweeteners lists the longest lead time (35 days). Packaging Pros has the best quality score (4.78). PO cancellations range from 16.1% (Packaging Pros) to 27.2% (Dairy Farms United); overall 22.9%. Planned PO lead time averages 29.4 days, which does not match the supplier table. No quality score is recorded for: Global Sweeteners.

Recommendation: Build a supplier scorecard, collect missing quality data and align planned lead times with actual performance.

5. Shipment delays are network-wide, not tied to one carrier or warehouse

About 25.1% of shipments are marked Delayed. The rate ranges only from 25.0% to 25.9% across carriers and 23.5% to 25.9% across origin warehouses.

Recommendation: Capture actual delivery dates so delay causes can be measured properly.

6. Demand is stable with modest growth

Average monthly units demanded: 2024: 12,803, 2025: 13,468, 2026: 13,631 (+6.5% from first to last year). Product shares range from 23.1% to 26.6%, led by Enchanted Milk Chocolate.

Recommendation: A steady reorder policy fits this demand; large seasonal buffers are not needed.

7. Conclusion

The analysis shows a stable demand pattern with the main risks in stock placement, supplier management and delivery tracking rather than in demand volatility. Priorities are: rebalance stock of the highest-value material, build a supplier scorecard, and record real delivery dates so delay causes can be measured.

Data limitations

- The dataset is synthetic (generated), so patterns are weaker than in real operational data.
- Shipment dates are not linked to order dates, and there are no delivery dates; delays use shipment status only.
- All stock movements are inbound, so inventory turnover cannot be calculated.
- Only a few suppliers, materials and warehouses exist, so rankings are indicative.
- The last month of sales data is partial and was excluded from monthly averages.